

Artificial Intelligence (AI2002)

Quiz # 01

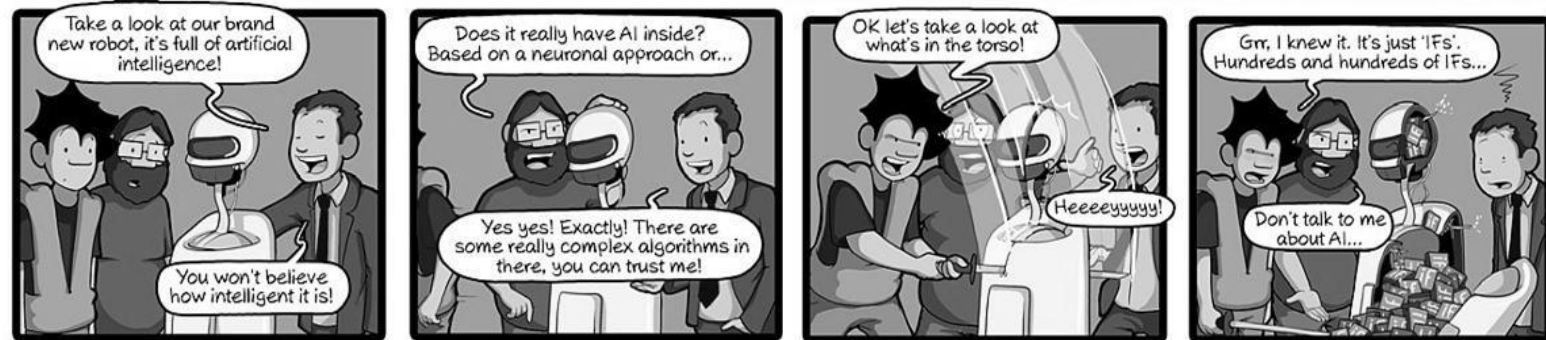
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Student Signature _____

Q1. Consider the comic below and answer the questions that follow: -

[8]



- A. Which dimension of AI definitions best matches the context in the above comic? Give strong reasons.
- B. What type of Agent did the robot turn out to be? Why were the guys disappointed to find many IFs? Isn't it AI?.
- C. Give PEAS for solving a jigsaw puzzle.
- D. Which of the categories of Nature of environments does solving jigsaw puzzles belong to? Cover all the characteristics of the task environment. (HINT: fully vs partially observable, deterministic vs nondeterministic etc.)

CLO2: To identify and relate methods of search and practically apply the corresponding techniques.

Q2. Suppose you have a transportation network with 10 cities represented by the adjacency matrix below. Each connection between cities has a specific weight (representing distance in kilometers). The goal is to analyze paths between City A and City J using different search algorithms.

[7]

	A	B	C	D	E	F	G	H	I	J
A	0	2	4	0	0	0	0	0	0	0
B	2	0	0	7	0	3	0	0	0	0
C	4	0	0	1	5	0	0	0	0	0
D	0	7	1	0	0	0	2	6	0	0
E	0	0	5	0	0	0	0	0	4	0
F	0	3	0	0	0	0	8	0	0	5
G	0	0	0	2	0	8	0	0	0	3
H	0	0	0	6	0	0	0	0	1	0
I	0	0	0	0	4	0	0	1	0	2
J	0	0	0	0	0	5	3	0	2	0

- a. Using BFS, what is the shortest path from City A to City J (considering unweighted paths)?
- b. Using DFS, find the longest path from City A to City J without revisiting any city.
- c. Using the Greedy Algorithm (always choose the nearest next city), what path would you find from City A to City J? Is this the optimal path? Justify your answer.